| Table 9.1 | | | |
|---|---|---|---|
| **Physical quantity** | **Ground state** | **First excited state** | **Second excited state** |
| Radius ($r_n \propto n^2$) | 0.529 Å | 2.116 Å | 4.761 Å |
| Velocity ($v_n \propto n^{-1}$) | $2.19 \times 10^6$ m s$^{-1}$ | $1.095 \times 10^6$ m s$^{-1}$ | $0.73 \times 10^6$ m s$^{-1}$ |
| Total Energy ($E_n \propto n^{-2}$) | –13.6 *e*V | –3.4 *e*V | –1.51 *e*V |

**energy required to remove an electron from the ground state of an atom to the outer most orbit *(E = 0* for *n*→∞*)* is known as first ionization energy (13.6 *e*V).** Then, the hydrogen atom is said to be in ionized state or simply called as hydrogen ion, denoted by $H^+$. If we supply more energy than the ionization energy, the excess energy appear as the kinetic energy of the free electron.

**Ionization potential is defined as ionization energy per unit charge.**

$$V_{\text{ionization}} = \frac{1}{e} E_{\text{ionization}} = \frac{13.6}{n^2} Z^2 \, V$$

Thus, for a hydrogen atom (Z=1), the ionization potential is

$$V = \frac{13.6}{n^2} \text{ volt}$$

The radius, velocity and total energy in ground state, first excited state and second excited state are given in Table 9.1.

## EXAMPLE 9.5

Suppose the energy of an electron in hydrogen–like atom is given as $E_n = -\frac{54.4}{n^2}$ *e*V where $n \in \mathbb{N}$. Calculate the following:

(a) Sketch the energy levels for this atom and compute its atomic number.

(b) If the atom is in ground state, compute its first excitation potential and also its ionization potential.

(c) When a photon with energy 42 *e*V and another photon with energy 51 *e*V are made to collide with this atom, does this atom absorb these photons?

(d) Determine the radius of its first Bohr orbit.

(e) Calculate the kinetic and potential energies of electron in the ground state.

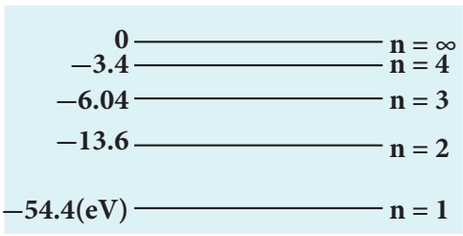
### *Solutions*

(a) Given that

$$E_n = -\frac{54.4}{n^2} \text{ eV}$$

For $n$ = 1, the ground state energy $E_1$ = –54.4 *e*V and for $n$ = 2, $E_2$ = –13.6 *e*V. Similarly, $E_3$ = –6.04 *e*V, $E_4$ = –3.4 *e*V and so on.

For large value of principal quantum number – that is, $n = \infty$, we get $E_\infty$ = 0 *e*V.




(b) For a hydrogen-like atom, ground state energy is

$$E_1 = -\frac{13.6}{n^2} Z^2 \text{ eV}$$

where $Z$ is the atomic number. Hence, comparing this energy with given energy, we get, $-13.6\,Z^2 = -54.4 \Rightarrow Z = \pm 2$. Since, atomic number cannot be negative number, $Z = 2$.



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